PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Misuse of medication in adult substance misuse services: a systematic review protocol
AUTHORS	Gittins, Rosalind; Missen, Louise; Maidment, Ian

VERSION 1 – REVIEW

REVIEWER	Chiappini, Stefania University of Hertfordshire Health and Human Sciences Research Institute
REVIEW RETURNED	17-Dec-2020

GENERAL COMMENTS	The systematic review protocol proposed by authors focuses on the misuse of both over the counter and prescription medications, which is an issue having an enormous impact upon physical and mental health due to drug interactions, overdoses and drug-related deaths. The protocol is interesting, well-written and supported by a strong methodology. However, please consider some major remarks the manuscript might benefit from: -the Introduction should be structured in paragraphs. Moreover please consider the following notes: Pag. 5 lines 4-6: among all, Z-drugs and quetiapine should be mentioned. Pag. 5 line 14: drug interactions should be considered, in fact alcohol and Illicit drugs are normally concomitantly abused, e.g. promethazine has normally misused together with alcohol and codeine-containing syrups with recreational purposes. It would be interesting to add a paragraph regarding how these drugs might be obtained. The Internet has for sure a role. Moreover, authors should specify if there are other ways users might obtain drugs, e.g. 'Doctor shopping' is a common reported phenomenon. Is there an illicit market of drugs? which drugs are mostly involved? Page 6, line 1: Authors mentioned the typical profile of misusers. Please give indications regarding this. An important point is how the misuse of medication might be studied: data regrading presciptions and post marketing and pharmacovigilance studies might contribute to understand these issues. Finally, several papers have been already published regarding this topic. Please consider the following refs: Assessing The 2004-2018 Fentanyl Misusing Issues Reported To An International Range Of Adverse Reporting Systems Chiappini S, Schifano F, Guirguis A, Corkery JM. Front. Pharmacol. 2019; 10:46. Doi: 10.3389/fphar.2019.00046

	Is there such a thing as a 'lope' dope? Analysis of loperamide-related European Medicines Agency (EMA) pharmacovigilance database reports. Schifano F, Chiappini S. PLoS ONE 2018; 13(10): e0204443. Doi: 10.1371/journal.pone.0204443. Recent changes in drug abuse; prescribing drug abuse in the context of Novel Psychoactive Substances (NPS) consumption: a systematic review. Schifano F, Chiappini S, Corkery J, Guirguis A. Brain Sci. 2018, 8(4), 73. Doi:10.3390/brainsci8040073. There is a potential of misuse for venlafaxine and bupropion? Schifano F, Chiappini S. Front. Pharmacol. 2018; 9:239. Doi: 10.3389/fphar.2018.0023921. β-2 agonists as misusing Drugs? Assessment of both Clenbuterol- and Salbutamol-related European Medicines Agency (EMA) Pharmacovigilance Database Reports. Milano G, Chiappini S, Mattioli F, Martelli A, Schifano F. Basic Clin Pharmacol Toxicol. 2018 Mar 2. Doi: 10.1111/bcpt.12991. Is there a potential of misuse for quetiapine? Literature review and analysis of the European Medicines Agency/EMA Adverse Drug Reactions' database. Chiappini S, Schifano F. J Clin Psychopharmacol. 2018 Feb;38(1):72-79. Doi: 10.1097/JCP.0000000000000814. A decade of gabapentinoid misuse: an analysis of the European Medicines Agency's 'suspected adverse drug reactions' database. Assessment of gabapentinoid misuse/dependence as reported to the EMA. Chiappini S, Schifano F. CNS drugs 2016, vol.30, n.5, pagg. 1-8. Doi: 10.1007/s40263-016-0359-y. Focus on Clozapine Withdrawal- and Misuse-Related Cases as Reported to the European Medicines Agency (EMA) Pharmacovigilance Database. Chiappini S, Schifano F, Corkery JM, Guirguis A. Brain Sci. 2020, 10(2), 105. Doi:10.3390/brainsci10020105. Pregabalin: a range of misuse-related unanswered questions. Schifano F, Chiappini S. CNS Neuroscience and Therapeutics 2019. Doi: 10.1111/cns.13115. An insight into Z-drugs abuse and dependence: an examination of reports to the European Medicines Agency (EMA) database of suspected Adverse Drug Reactions (ADR). Schifano F, Chiappini S, Corkery JM, Guirguis A. IJNP 2019, pyz007. Doi: 10.1093/ijnp/pyz007. Understanding the use of prescription and OTC drugs in obtaining highs and the pharmacist role in preventing abuse. Chiappini S, Guirguis A, Corkery J, Schifano F.

	The Pharmaceutical Journal Vol.305, No 7943, online (https://www.pharmaceutical-journal.com/research/perspective-article/misuse-of-prescription-and-over-the-counter-drugs-to-obtain-illicit-highs-how-pharmacists-can-prevent-abuse/20208538.article?firstPass=false) DOI: 10.1211/PJ.2020.20208538. What about 'pharming'? Issues regarding the misuse/abuse of prescription and over-the-counter drugs. Chiappini S, Schifano F. Brain Sci. 2020 Oct; 10(10): 736. Beyond the PURPLE DRANK. Study of promethazine abuse according to the European Medicines Agency (EMA) Adverse Drug Reactions (ADR) reports. Chiappini S, Schifano F, Corkery JM, Guirguis A. J Psychopharmacology, IN PRESS. The misuse of prescription drugs in the digital era: pros and cons. Chiappini S, Guirguis A, Corkery JM, Schifano F. Abstract for the RPS Science & Research Summit (June 2020). COVID-19: The hidden impact on mental health and drug addiction. Chiappini S, Guirguis A, John A, Corkery JM, Schifano F. Frontiers in Psychiatry. Accepted July 20th, 2020. Doi:10.3389/fpsyt.2020.00767. 'Z-trip'? A comprehensive overview and and a case-series of Zolpidem misuse. Orsolini L, Chiappini S, Grandinetti P, Bruschi A, Testa R, Provenzano A, De Berardis D. Clinical Psychopharmacology and Neuroscience. DOI: 10.9758/cpn.2020.18.2.1.

REVIEWER	Cooper, Richard University of Sheffield, ScHARR
REVIEW RETURNED	12-Jan-2021

GENERAL COMMENTS	This is a helpful initial presentation of a proposed SR protocol but does require some important changes before being suitable for consideration and publication which I hope the authors are able to make and deliver on the funding from CMHP and PRUK. Firstly there is a numbered list of specific aspects that require ley review, and this is followed by more minor in text comments. 1) Important need to define 'misuse' and related terms, particularly as 'abuse' and 'diversion' are referred to; the latter is important but could be viewed as a separate issue, and then brings in issues of whether the substance being used problematically and resulting in formal treatment was obtained legally (Rx or OTC) or if diverted. There is a lot of overlap in the formal drug trwatment data (particularly benzo but also opioid) with co-presentation of illicit substance misuse and diversion is more likely in this population. Also, is there a need to consider terms such as 'dependence' which UK formal drug treatment services use (and the PHE report and Marsden et al 2019 paper similarly did), but also addiction and see later comment but other MeSH terms.

	2) see detailed comments below but felt some of the supporting arguments could be refined slightly more to give more of a robust defence based on evidence. 3) clarification of SMS - in the UK this would be formal drug and alcohol treatment services (eg DAAT, BUT cf GP attempts to treat, or pain clinics). 6/29 refers to 'any setting' which is actually fine and I'd argue we do need a comprehensive holistic understanding of this issues BUT arguably an SMS is a more specific activity, and a better definition is therefore needed. So two options: a) keep SMS but make sure you don't include other support interventions, or b) define and exclude non-formal SMS. 4) Eligibility criteria need tidying-up a bit as currently seem to be non-equivalent and not precise enough and also linked to 1) and 3). 5) No time period is included which might be useful. 6) Scopus and WoS do not seem comprehensive enough - suggest the authors re-check existing health-related databases (eg Pubmed or OVID Medline, but why no PsychInfo or Cochrane also)? Check the PHE/Marsden et al 2019 searches used. 7) I don't think the current search terms and strategy will be sufficiently robust; I realise this is a protocol but I don't think what is being proposed would capture all the relevant literature. 8) The MMAT is proposed to assess risk of bias which is appropriate if mixed methods studies are anticipated but it might be wiser to consider more standard critical appraisal tools (CASP etc) for mono-method studies (which are also more common - ie quant or qual only). 9) Linked to 8) why is there only reference to a qualitative synthesis if quant and mixed methods are anticipated; it would not seem appropriate to report on the quality and findings from quant studies only in a qualitative way; meta-analysis is often not possible (heterogeneity etc) but still worth reporting quant as numerical descriptives and good that the authors consider this. 10) NDTMS data would arguably also provide an answer to some of the aims and also wondered as a final point if this needed to be considered as part of the defence of the systematic review methods. Minor comments 4/3 also mention z-drugs and not sure if it is helpful to specifically refer to gregabalin and diazepam and better as therapeutic classes arguably (and to mention benzodiazepine for example, and gabapentinoid perhaps?) 5/5 'such as' not 'including'? 5/9 blood-borne (https://wwwnc.cdc.gov/eid/page/punctuation) and explain QT but also very specific so perhaps review. 5/22 is Buscopan brand name necessary? 5/23 supra-doses - is this a common term? 5/30 reference not ideal as the ACMD report is on diversion. 5/39 and 6/4 - agree the PHE review is an important one and the Marsden associated publication which is oddly omitted (https://www.thelancet.com/journals/lanpsy/article/PIIS2215-0366(19)30331-1/fulltext) BUT not sure we can term it landmark (yet) so remove that adjective possibly. 5/49 - remove 'latest' as will be anachronistic in the future. 5/52 - the argument based on ref 42 (Bohnert et al) needs reviewing as it was not those who access SMS for POM/OTC but any addiction (actually 30.2% alcohol, 19.4% heroin, 16.0% cocaine, 9.7% marijuana, 4.0% other opiates, and 20.8% other or missing) and the study was a survey of whether they used an opioid appropriately or abused it).

	5/58 'Evidence in the US suggests that' needed to support claim as only based on this country and not actually generalisable. 5/58 Also in this section, the argument may be a tautology - when POM/OTC misusers engage in SMS then outcomes are good, therefore analysis of misusers' demographics is important. This would suggest ideally we should compare those who engage and those who do not, and not, as planned here, only those in SMS. 5/59 and 6/3 'demographics' clauses repeat 6/14 'used [...] when writing our report' - do you mean your protocol? The report is later, or do you mean you anticipate using 6/26 eligibility criteria grammatically not precise enough and although helpfully seems to follow some PICOS, needs to for example state more explicitly 'adult participants (18 years or over)' but also as per comments above, key terms and definitions need to be considered and if misuse only is used or might MeSH search be appropriate and other synonyms. I hope these help and result in a robust search that can provide important evidence on this topic, which I think in time will complement other research.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1	Author responses
-the Introduction should be structured in paragraphs.	The introduction is structured in paragraphs. Unclear what is meant by this.
Pag. 5 lines 4-6: among all, Z-drugs and quetiapine should be mentioned.	Added specific reference to these medicines to paragraph 5 in the Introduction. Other medicines including zopiclone (a z-drug) also mentioned in the Introduction paragraph 6. However, this specific point relates to the medication most commonly associated with drug related deaths which are of most concern, so z-drugs and quetiapine are not specifically mentioned here, though are broadly covered by their effects being listed.
Pag. 5 line 14: drug interactions should be considered, in fact alcohol and Illicit drugs are normally concomitantly abused, e.g. promethazine has normally misused together with alcohol and codeine-containing syrups with recreational purposes.	Thanks for highlighting this and we completely agree. Drug interactions including alcohol and illicit drugs are mentioned in the first and second paragraphs of the Introduction. Promethazine/codeine syrups aka ‘lean’ is mentioned in the Introduction paragraph 4.
It would be interesting to add a paragraph regarding how these drugs might be obtained. The Internet has for sure a role. Moreover, authors should specify if there are other ways users might obtain drugs, e.g. 'Doctor shopping' is a common reported phenomenon. Is there an illicit	Added to the Introduction as suggested. The purpose of this review is to hopefully identify some of the questions the reviewer has raised (e.g. which drugs are mostly involved).

market of drugs? which drugs are mostly involved?	
Page 6, line 1: Authors mentioned the typical profile of misusers. Please give indications regarding this.	Rephrased and updated accordingly as suggested.
An important point is how the misuse of medication might be studied: data regrading presciptions and post marketing and pharmacovigilance studies might contribute to understand these issues.	Alternative methodologies are considered in the Introduction paragraph 5: post-marketing surveillance and pharmacovigilance studies have been added with further addition of the reviewer citation as suggested in point below. Furthermore, such reports will be included in the systematic review if they meet the inclusion criteria.
Finally, several papers have been already published regarding this topic. Please consider the following refs: ….	Thank you for drawing attention to this important literature. These references have been noted and we have added to the citations from this body of work. Subsequently the reference list has been re-numbered. Further papers will be included when undertaking the systematic review if they meet the inclusion criteria.
Reviewer: 2	Author responses
1) Important need to define 'misuse' and related terms, particularly as 'abuse' and 'diversion' are referred to; the latter is important but could be viewed as a separate issue, and then brings in issues of whether the substance being used problematically and resulting in formal treatment was obtained legally (Rx or OTC) or if diverted. There is a lot of overlap in the formal drug trwatment data (particularly benzo but also opioid) with co-presentation of illicit substance misuse and diversion is more likely in this population. Also, is there a need to consider terms such as 'dependence' which UK formal drug treatment services use (and the PHE report and Marsden et al 2019 paper similarly did), but also addiction and see later comment but other MeSH terms.	Thanks for highlighting this. Additional detail added as requested to Introduction paragraph one. Numerous minor changes subsequently made to the text in line with this recommendation.
2) see detailed comments below but felt some of the supporting arguments could be refined slightly more to give more of a robust defence based on evidence.	Information added – see below for further info.

3) clarification of SMS - in the UK this would be formal drug and alcohol treatment services (eg DAAT, BUT cf GP attempts to treat, or pain clinics). 6/29 refers to 'any setting' which is actually fine and I'd argue we do need a comprehensive holistic understanding of this issues BUT arguably an SMS is a more specific activity, and a better definition is therefore needed. So two options: a) keep SMS but make sure you don't include other support interventions, or b) define and exclude non-formal SMS.	Agree, and this is made more difficult in part because of how services are commissioned e.g. GP shared care provision. Anticipate publications may be variable in how well this is defined and perhaps different in publications from non-UK systems. Overall, we have amended the text in line with the recommendations and went for 'option a' i.e. structured treatment to provide greater clarity and manuscript updated accordingly – notably changes made to the Eligibility Criteria.
4) Eligibility criteria need tidying-up a bit as currently seem to be non-equivalent and not precise enough and also linked to 1) and 3).	Updated accordingly.
5) No time period is included which might be useful.	Added to abstract and Methods as suggested.
6) Scopus and WoS do not seem comprehensive enough - suggest the authors re-check existing health-related databases (eg Pubmed or OVID Medline, but why no PsychInfo or Cochrane also)? Check the PHE/Marsden et al 2019 searches used.	OVID Medline, Cochrane and Pubmed have been added as suggested. The Abstract and Methods sections and Reference list have been updated to reflect this. Aston University library staff have been consulted to confirm the access required and the suitability of the databases selected, and the resulting systematic review will note the limitation that the University does not currently have access to every possible desirable database.
7) I don't think the current search terms and strategy will be sufficiently robust; I realise this is a protocol but I don't think what is being proposed would capture all the relevant literature.	Following this valuable point being raised, Aston University library staff have been consulted to confirm the suitability of the proposed search strategy which has been further refined. A MeSH search and further detail has been added to Methods section. Piloting has indicated that this search methodology is likely to identify appropriate numbers and relevant literature. An exemplar for OVID Medline has been provided in a Supplementary File. The Acknowledgements section has also been updated to reflect the support provided.
8) The MMAT is proposed to assess risk of bias which is appropriate if mixed methods studies are anticipated but it might be wiser to consider more standard critical appraisal tools (CASP etc) for mono-method studies (which are also more common - ie quant or qual only).	Mixed methods are anticipated. One tool has been chosen to help standardise the assessment process. This will be noted as a potential limitation should any mono-method studies be identified.

9) Linked to 8) why is there only reference to a qualitative synthesis if quant and mixed methods are anticipated; it would not seem appropriate to report on the quality and findings from quant studies only in a qualitative way; meta-analysis is often not possible (heterogeneity etc) but still worth reporting quant as numerical descriptives and good that the authors consider this.	Thanks for this comment; we completely agree that meta-analysis is not anticipated due to the heterogeneity of the data. However, 'numerical descriptives' will be captured as a narrative and we have clarified this in the Methods section.
10) NDTMS data would arguably also provide an answer to some of the aims and also wondered as a final point if this needed to be considered as part of the defence of the systematic review methods.	The problem with NDTMS data being that it does not report on all substances/medicines used and requires disclosure. Additionally, NDTMS does not apply outside of the UK and is highly unlikely to be included in publications.
4/3 also mention z-drugs and not sure if it is helpful to specifically refer to gregabalin and diazepam and better as therapeutic classes arguably (and to mention benzodiazepine for example, and gabapentinoid perhaps?)	This point relates to the medication most commonly associated with drug related deaths which are of most concern so z-drugs not mentioned here specifically though broadly captured by effects listed. Broader medicine group stated as in the literature they are referred to as such – and in the case of diazepam others e.g. alprazolam are of greater concern so therefore left as medicines group. However, added specific medicines to paragraph 5 of the introduction to simultaneously cover a point raised by reviewer 1.
5/5 'such as' not 'including'?	Changed as suggested.
5/9 blood-borne (https://wwwnc.cdc.gov/eid/page/punctuation) and explain QT but also very specific so perhaps review.	Changed as suggested. Added detail about QT: this is specific but of notable clinical significance.
5/22 is Buscopan brand name necessary?	Brand used as this is the medication that the alerts related to and is the only version used in the UK.
5/23 supra-doses - is this a common term?	Thanks for raising this – we have updated the wording to make explicit that what is meant is 'significantly high' doses.
5/30 reference not ideal as the ACMD report is on diversion.	Noted, though the report comments upon diversion and misuse, so the text has been amended accordingly to differentiate this as recommended.
5/39 and 6/4 - agree the PHE review is an important one and the Marsden associated publication which is oddly omitted (https://www.thelancet.com/journals/lanpsy/article/PIIS2215-0366(19)30331-1/fulltext) BUT not sure we can term it landmark (yet) so remove that adjective possibly.	Added the associated Marsden publication. All sequential reference numbering consequently updated. The second mention of the word 'Landmark' has been removed, but the first one left in as this is how the government described the review and has been referred to as such in numerous publications/media commentaries – e.g.: • https://healthmedia.blog.gov.uk/2018/01/26/government-launches-landmark-review-into-prescription-drug-addiction/ • https://www.independent.co.uk/news/health/prescription-drug-addiction-uk-review-health-department-jeremy-hunt-care-a8174706.html

	• http://www.pharmatimes.com/news/phe_to_review_prescription_drug_addition_in_the_uk_1218519 • https://www.bmj.com/content/366/bmj.l5497
5/49 - remove 'latest' as will be anachronistic in the future.	Removed as suggested.
5/52 - the argument based on ref 42 (Bohnert et al) needs reviewing as it was not those who access SMS for POM/OTC but any addiction (actually 30.2% alcohol, 19.4% heroin, 16.0% cocaine, 9.7% marijuana, 4.0% other opiates, and 20.8% other or missing) and the study was a survey of whether they used an opioid appropriately or abused it).	Reference changed and wording updated as suggested.
5/58 'Evidence in the US suggests that' needed to support claim as only based on this country and not actually generalisable.	Amended and rewritten to reflect this important point.
5/58 Also in this section, the argument may be a tautology - when POM/OTC misusers engage in SMS then outcomes are good, therefore analysis of misusers' demographics is important. This would suggest ideally we should compare those who engage and those who do not, and not, as planned here, only those in SMS.	Amended wording.
5/59 and 6/3 'c' clauses repeat	Removed second mention.
6/14 'used [...] when writing our report' - do you mean your protocol? The report is later, or do you mean you anticipate using	Amended wording.
6/26 eligibility criteria grammatically not precise enough and although helpfully seems to follow some PICOS, needs to for example state more explicitly 'adult participants (18 years or over)' but also as per comments above, key terms and definitions need to be considered and if misuse only is used or might MeSH search be appropriate and other synonyms.	Changes made as suggested (see also response relating to this point above).

VERSION 2 – REVIEW

REVIEWER	Chiappini, Stefania

	University of Hertfordshire Health and Human Sciences Research Institute
REVIEW RETURNED	22-Feb-2021

GENERAL COMMENTS	Thank you for the revisions, the manuscript has really improved.

REVIEWER	Cooper, Richard University of Sheffield, ScHARR
REVIEW RETURNED	13-Mar-2021

GENERAL COMMENTS	Revised paper addresses many issues identified in the first review